LESSON 5

DIELECTRIC SLAB WAVEGUIDE

5.1 Introduction

The mathematical treatment of dielectric slab waveguide is similar to those for a rectangular waveguide only that the boundary condition considers only two surfaces.

5.2 Dielectric slab waveguides

-Guided waveguide modes can be supported in a sufficiently thick dielectric slab surrounded by cladding regions of lower refractive index.

-Consider Fig. 5.1 below.

The slab is assumed to be of infinite extent in the ± y directions so that the fields do not show dependence on y (i.e. $\frac{\partial E}{\partial y} = \frac{\partial H}{\partial y} = 0$).

-Propagation is to be in the +z direction and each region is assumed to be lossless (i.e $\gamma = i\beta$ or $\alpha = 0$).

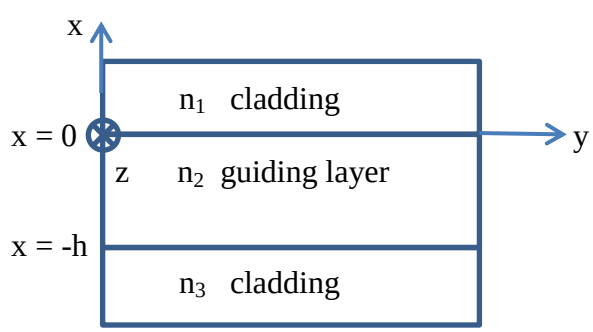

Figure 5.1 Dielectric waveguide

5.3 General equations for propagation

In order to get the general equation governing propagation of electromagnetic wave we use Maxwell's modified equations since the waveguide is hollow ($\rho = J = 0$) . These are

$$\nabla \cdot \vec{E} = 0 \tag{5.1}$$

$$\nabla \cdot \vec{B} = 0 \tag{5.2}$$

$$\nabla \mathrm{x} \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -i\omega\mu\vec{H} \tag{5.3}$$

$$\nabla x\vec{H} = \frac{\partial \vec{D}}{\partial t} = i\omega\varepsilon\vec{E} \tag{5.4}$$

Taking the curl of eq. 5.3 gives

$$\nabla x\nabla x\vec{E} = -i\omega\mu(\nabla x\vec{H}) \tag{5.6}$$

Substituting eq. 5.3 into eq. 5.6 gives

$$\nabla x\nabla x\vec{E} = \omega^2\mu\varepsilon\vec{E} \tag{5.7}$$

Applying the identity $\nabla x\nabla x\vec{M} = \nabla.\left(\nabla \cdot \vec{M}\right) - \nabla^2\vec{M}$ to the left hand side of eq. 5.7 gives

$$\nabla.\left(\nabla \cdot \vec{E}\right) - \nabla^2\vec{E} = \omega^2\mu\varepsilon\vec{E} \tag{5.8}$$

Substituting eq. 5.1 into eq. 5.8 gives

$$\nabla^2\vec{E} + \omega^2\mu\varepsilon\vec{E} = 0 \tag{5.9}$$

Similarly taking the curl of eq. 5.4 and proceed as before gives

$$\nabla^2\vec{H} + \omega^2\mu\varepsilon\vec{H} = 0 \tag{5.10}$$

Using the Laplacian $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ assuming $\frac{\partial^2}{\partial y^2} = 0$, gives

$$\nabla^2 E_z + \omega^2\mu\varepsilon E_z = k^2 E_z \tag{5.115j}$$

and

$$\nabla^2 H_z + \omega^2\mu\varepsilon H_z = k^2 H_z \tag{5.126k}$$

apply in both cladding and guiding regions.

5.4 Guiding region n_2

-Over the guiding layer ($-h \leq x \leq 0$) the x-variation of solution of eqn. 5.11 is

$$E_z(x) = Asin(q_2x) + Bcos(q_2x) \tag{5.13}$$

where A and B are constants.

Substituting $\omega^2\mu\varepsilon - k^2 = k_o^2 n_i^2 - k^2$ in eq. 5.11 gives

$$\nabla^2 E_z + [k_o^2 n_2^2 - k^2]E_z = 0 \tag{5.14}$$

Substituting eq. 5.137a into 5.147b, assuming no variation in y-direction $\frac{\partial^2}{\partial y^2} = 0$ gives

$$\frac{\partial^2}{\partial x^2}[Asin(q_2x) + Bcos(q_2x)] + [k_o^2n_2^2 - k^2][Asin(q_2x) + Bcos(q_2x)] = 0$$

$$[-Aq_2^2sin(q_2x) - q_2^2Bcos(q_2x)] + [k_o^2n_2^2 - k^2][Asin(q_2x) + Bcos(q_2x)] = 0$$

$$q_2^2 = k_o^2n_2^2 - k^2 \qquad \text{for guiding region} \qquad (5.157c)$$

5.5 Upper cladding region n_1

The amplitude of the wave decays exponentially into the upper cladding region. Here the solution is

$$E_z = Ce^{-p_1x} \qquad \text{for upper cladding region } x \geq 0 \qquad (5.168)$$

Where C is a constant. Substituting eq. 5.16 into eq. 5.115j gives

$$\frac{\partial^2}{\partial x^2}[Ce^{-p_1x}] + [k_o^2n_1^2 - k^2][Ce^{-p_1x}] = 0$$

$$p_1^2 = k^2 - k_o^2n_1^2 \qquad \text{for upper cladding region} \qquad (5.17) \quad 5.10$$

5.6 Lower cladding region n_3

The amplitude of the wave decays exponentially into the lower cladding region. Here the solution is

$$E_z = De^{-p_3(x+h)} \qquad \text{for lower cladding region } x \leq -h \qquad (5.18) \quad 5.9$$

where D is a constant. Substituting eq. 5.18 into eq. 5.115j gives

$$\frac{\partial^2}{\partial x^2}\left[De^{-p_3(x+h)}\right] + [k_o^2n_3^2 - k^2]\left[De^{-p_3(x+h)}\right] = 0$$

$$p_3^2 = k^2 - k_o^2n_3^2 \qquad \text{for lower cladding region} \qquad (5.19)$$

5.7 Cut-off condition

For propagation to take place $p_3 = \sqrt{k^2 - k_o^2n_3^2}$ must be real which from eq. 5.19, demands that $k \geq k_on_3$.

This is the cut-off condition.

5.8 General expressions for E_x, E_y, H_x and H_y field components

The restriction, $\frac{\partial}{\partial y} = 0$, allows fields to be

$$E_y = \frac{i}{k_c^2}\left(-k\frac{\partial E_z}{\partial y} + \omega\mu\frac{\partial H_z}{\partial x}\right) = \frac{i\omega\mu}{k_c^2}\frac{\partial H_z}{\partial x} \quad (5.20)$$

$$H_x = \frac{i}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial y} - k\frac{\partial H_z}{\partial x}\right) = -\frac{ik}{k_c^2}\frac{\partial H_z}{\partial x} \quad (5.21)$$

$$E_x = -\frac{i}{k_c^2}\left(k\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right) = -\frac{ik}{k_c^2}\frac{\partial E_z}{\partial x} \quad (5.22)$$

$$H_y = -\frac{i}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial x} + k\frac{\partial H_z}{\partial y}\right) = -\frac{i\omega\varepsilon}{k_c^2}\frac{\partial E_z}{\partial x} \quad (5.23)$$

where $k_c^2 = \omega^2\mu\varepsilon - k^2$.

Because the three regions have different refractive indices, k_c needs to be expressed in terms of refractive index, giving

$k_c^2 = \omega^2\mu\varepsilon - k^2 = k_o^2c^2\frac{1}{v^2} - k^2 = k_o^2n_i^2 - k^2$ and i = 1,2,3,4, i.e. regions 1,2,3

Because the three regions have different permittivities and refractive indices a relation between them need to be found.

$n = \frac{c}{v} = \frac{\sqrt{\varepsilon_r\varepsilon_o\mu_r\mu_o}}{\sqrt{\varepsilon_o\mu_o}} = \frac{\sqrt{\varepsilon_r 1\varepsilon_o\mu_o}}{\sqrt{\varepsilon_o\mu_o}} = \sqrt{\varepsilon_r}$ or $\quad \varepsilon_r = n^2$ or $\quad \varepsilon_{ri} = n_i^2$

5.2.3 TM modes

-For the TM mode, since $H_z = 0$ and $E_z \neq 0$, field components are

$$E_y = \frac{i\omega\mu}{k_c^2}\frac{\partial H_z}{\partial x} = 0 \quad (5.20)$$

$$H_x = -\frac{ik}{k_c^2}\frac{\partial H_z}{\partial x} = 0 \quad (5.21)$$

$$E_x = -\frac{ik}{k_c^2}\frac{\partial E_z}{\partial x} \neq 0 \quad (5.22)$$

$$H_y = -\frac{i\omega\varepsilon}{k_c^2}\frac{\partial E_z}{\partial x} \neq 0 \quad (5.23)$$

There two tasks:

First we state the equations relating A, B, C and D are

$A = \frac{n_1^2q_2B}{n_2^2p_1}$, $C = B$ and $D = -Asin(q_2h) + Bcos(q_2h)$

Secondly, we need to get the solution for the general equation governing propagation for TM mode by using the interface between regions 2 and 3.

-Equating $H_y = -\frac{i\omega\varepsilon}{k_c^2}\frac{\partial E_z}{\partial x}$ between regions 2 and 3 at x = -h and substituting for $E_z(x) = Asin(q_2x) + Bcos(q_2x)$ in region 2 and $E_z(x) = De^{-p_3(x+h)}$ in region 3 gives

GUIDING REGION 2 $\Downarrow$ LOWER REGION 3 $\Downarrow$

$$-\frac{i\omega\varepsilon_2}{k_c^2}\frac{\partial}{\partial x}[Asin(q_2x) + Bcos(q_2x)] = -\frac{i\omega\varepsilon_3}{k_c^2}\frac{\partial}{\partial x}\left[De^{p_3(x+h)}\right]$$

$$\frac{i\omega\varepsilon_2}{k_c^2}[q_2Acos(q_2x) - q_2Bsin(q_2x)] = \frac{i\omega\varepsilon_3}{k_c^2}\left[p_3De^{p_3(x+h)}\right] \quad (5.24a)$$

Substituting for $D = -Asin(q_2h) + Bcos(q_2h)$ into RHS of eq. 5.24a, gives

$$\frac{i\omega\varepsilon_2 q_2}{k_c^2}[Acos(q_2x) - Bsin(q_2x)] = \frac{i\omega\varepsilon_3}{k_c^2}\left[p_3\{-Asin(q_2h) + Bcos(q_2h)\}e^{p_3(x+h)}\right] \quad (5.24b)$$

Substituting for $k_c^2 = k_o^2n_2^2 - k^2$ on LHS and $k_c^2 = k_o^2n_3^2 - k^2$ on the RHS

$$\frac{i\omega\varepsilon_2 q_2}{k_o^2n_2^2 - k^2}\left[Acos(q_2x) - Bsin(q_2x)\right] = \frac{i\omega\varepsilon_3}{k_o^2n_3^2 - k^2}\left[p_3\{-Asin(q_2h) + Bcos(q_2h)\}e^{p_3(x+h)}\right] \quad (5.24c)$$

Substituting for $q_2^2 = k_o^2n_2^2 - k^2$ on LHS and $-p_3^2 = k_o^2n_3^2 - k^2$ on the RHS of eq 5.24c

$$\frac{i\omega\varepsilon_2 q_2}{q_2^2}\left[Acos(q_2x) - Bsin(q_2x)\right] = -\frac{i\omega\varepsilon_3}{p_3^2}\left[p_3\{-Asin(q_2h) + Bcos(q_2h)\}e^{p_3(x+h)}\right] \quad (5.24d)$$

At x=-h, ,eq. 5.24d becomes

$$\frac{i\omega\varepsilon_2}{q_2}\left[Acos(-q_2h) - Bsin(-q_2h)\right] = -\frac{i\omega\varepsilon_3}{p_3}\left[\{-Asin(q_2h) + Bcos(q_2h)\}e^0\right] \quad (5.25a)$$

Assuming cos(-θ)=cosθ and sin(-θ)=-sinθ, eq. 5.25a becomes

$$\frac{i\omega\varepsilon_2}{q_2}\left[Acos(q_2h) + Bsin(q_2h)\right] = -\frac{i\omega\varepsilon_3}{p_3}\left[\{-Asin(q_2h) + Bcos(q_2h)\}\right]$$

Substituting $\varepsilon_2 = n_2^2$ on the LHS and $\varepsilon_3 = n_2^2$ on the RHS of eq. 5.25a we get

$$\frac{n_2^2}{q_2}\left[Acos\left(q_2h\right)+Bsin\left(q_2h\right)\right]=\frac{n_3^2}{-p_3}\left[-Asin(q_2h)+Bcos(q_2h)\right] \qquad (5.25b)$$

Substituting $A=\frac{n_1^2q_2B}{n_2^2p_1}$ into eq. 5.25b, we get

$$\frac{n_2^2}{q_2}\left[\left(\frac{n_1^2q_2B}{n_2^2p_1}\right)cos(q_2h)+Bsin(q_2h)\right]=\frac{n_3^2}{-p_3}\left[-\left(\frac{n_1^2q_2B}{n_2^2p_1}\right)sin(q_2h)+Bcos(q_2h)\right]$$

Dividing both sides by B gives

$$\frac{n_2^2}{q_2}\left[\left(\frac{n_1^2q_2}{n_2^2p_1}\right)cos(q_2h)+sin(q_2h)\right]=\frac{n_3^2}{-p_3}\left[-\left(\frac{n_1^2q_2}{n_2^2p_1}\right)sin(q_2h)+cos(q_2h)\right]$$

Dividing both sides by $cos(q_2h)$ gives

$$\frac{n_2^2}{q_2}\left[\left(\frac{n_1^2q_2}{n_2^2p_1}\right)+tan(q_2h)\right]=\frac{n_3^2}{-p_3}\left[-\left(\frac{n_1^2q_2}{n_2^2p_1}\right)tan(q_2h)+1\right]$$

$$\left[\left(\frac{n_1^2}{p_1}\right)+\frac{n_2^2}{q_2}tan(q_2h)\right]=\left[\left(\frac{n_1^2q_2n_3^2}{n_2^2p_1p_3}\right)tan(q_2h)-\frac{n_3^2}{p_3}\right]$$

Rearranging

$$\left(\frac{n_1^2q_2n_3^2}{n_2^2p_1p_3}\right)tan(q_2h)-\frac{n_2^2}{q_2}tan(q_2h)=\left(\frac{n_1^2}{p_1}\right)+\frac{n_3^2}{p_3} \qquad 5.26$$

If we let $n_1^2=\frac{n_2^2}{r_1}$ and $n_3^2=\frac{n_2^2}{r_3}$ we get

$$\left(\frac{\frac{n_2^2}{r_1}q_2\frac{n_2^2}{r_3}}{n_2^2p_1p_3}\right)tan(q_2h)-\frac{n_2^2}{q_2}tan(q_2h)=\left(\frac{\frac{n_2^2}{r_1}}{p_1}\right)+\frac{\frac{n_2^2}{r_3}}{p_3}$$

$$\left(\frac{n_2^2q_2}{r_1r_3p_1p_3}\right)tan(q_2h)-\frac{n_2^2}{q_2}tan(q_2h)=\left(\frac{n_2^2}{p_1r_1}\right)+\frac{n_2^2}{p_3r_3}$$

$$\left[\left(\frac{n_2^2q_2}{r_1r_3p_1p_3}\right)-\frac{n_2^2}{q_2}\right]tan(q_2h)=\left(\frac{n_2^2p_3r_3+n_2^2p_1r_1}{p_1r_1p_3r_3}\right)=\left(\frac{n_2^2[p_3r_3+p_1r_1]}{p_1r_1p_3r_3}\right)$$

$$\left[\left(\frac{n_2^2q_2}{r_1r_3p_1p_3}\right)-\frac{n_2^2}{q_2}\right]tan(q_2h)=\left(\frac{n_2^2p_3r_3+n_2^2p_1r_1}{p_1r_1p_3r_3}\right)=\left(\frac{n_2^2[p_3r_3+p_1r_1]}{p_1r_1p_3r_3}\right)$$

$$\left(\frac{n_2^2q_2^2-n_2^2[r_1r_3p_1p_3]}{r_1r_3p_1p_3q_2}\right)tan(q_2h)=\left(\frac{n_2^2[p_3r_3+p_1r_1]}{p_1r_1p_3r_3}\right)$$

$$\left(\frac{n_2^2(q_2^2-r_1r_3p_1p_3}{r_1r_3p_1p_3q_2}\right)tan(q_2h) = \left(\frac{n_2^2[p_3r_3+p_1r_1]}{p_1r_1p_3r_3}\right)$$

$$\left(\frac{q_2^2-r_1r_3p_1p_3}{r_1r_3p_1p_3q_2}\right)tan(q_2h) = \left(\frac{p_3r_3+p_1r_1}{p_1r_1p_3r_3}\right)$$

$$tan(q_2h) = \frac{\left(\frac{p_3r_3+p_1r_1}{p_1r_1p_3r_3}\right)}{\left(\frac{q_2^2-r_1r_3p_1p_3}{r_1r_3p_1p_3q_2}\right)} = \frac{q_2(p_3r_3+p_1r_1)}{q_2^2-r_1r_3p_1p_3} \qquad 5.27$$

This is the solution for TM mode. In general eq. 5.27b can be written cyclically as

$$\tan(q_2h - m\pi) = \frac{q_2(p_1r_1+p_3r_3)}{q_2^2-r_1r_3p_1p_3} \qquad (5.28)$$

where m = 0, 1, 2,… is the mode number. m stands for the number of times H_y crosses the x axis. The guided mode with m = 0 is called the fundamental mode and those with m = 1, 2, … are higher-order modes. In TM mode, the magnetic field (H_y) is perpendicular to the direction of propagation.

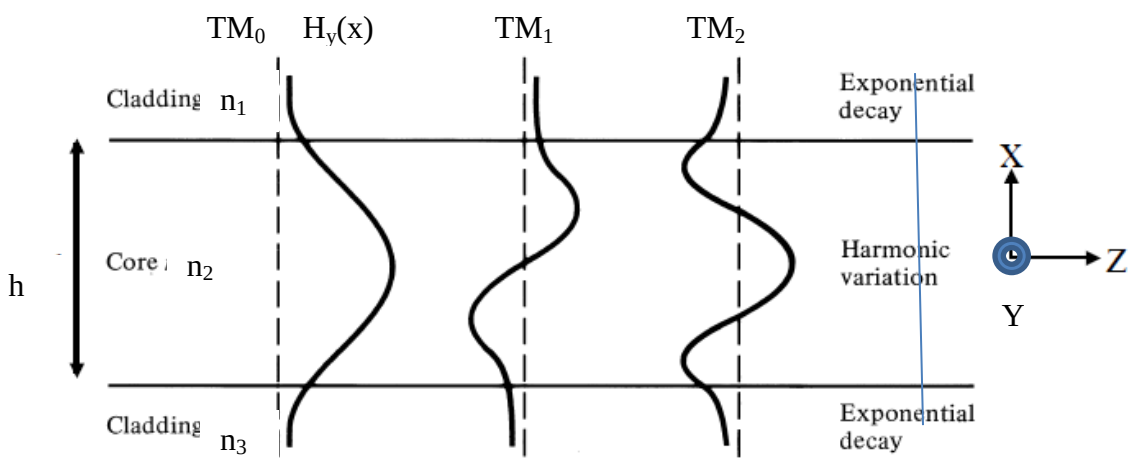

Fig. Field patterns for TM_0, TM_1 and TM_2 modes in dielectric slab waveguide.

5.2.3 TE modes

-For the TE mode, since $E_z = 0$ and $H_z \neq 0$, field components are

$$E_y = \frac{i\omega\mu}{k_c^2}\frac{\partial H_z}{\partial x} \neq 0 \qquad (5.20)$$

$$H_x = -\frac{ik}{k_c^2}\frac{\partial H_z}{\partial x} \neq 0 \qquad (5.21)$$

$$E_x = -\frac{i}{k_c^2} = -\frac{ik}{k_c^2}\frac{\partial E_z}{\partial x} = 0 \qquad (5.22)$$

$$H_y = -\frac{i}{k_c^2} = -\frac{i\omega\varepsilon}{k_c^2}\frac{\partial E_z}{\partial x} = 0 \qquad (5.23)$$

There are two tasks

First: we state the equations relating A, B and D given by

$$A = \frac{q_2 B}{p_1} \qquad \text{and} \qquad D = -Asin(q_2 h) + Bcos(q_2 h)$$

Second: we need to get the solution to the general equation governing propagation for TE mode by using the interface between regions 2 and 3.

Equating $H_x = -\frac{ik}{k_c^2}\frac{\partial H_z}{\partial x}$ between regions 2 and 3 at x=-h. Substituting for $H_z(x) = Asin(q_2 x) + Bcos(q_2 x)$ in region 2 and $H_z(x) = De^{-p_3(x+h)}$ in region 3 giving

GUIDING REGION 2 $\Downarrow$ LOWER REGION 3 $\Downarrow$

$$\frac{ik}{k_c^2}\frac{\partial}{\partial x}[Asin(q_2 x) + Bcos(q_2 x)] = \frac{ik}{k_c^2}\frac{\partial}{\partial x}\left[De^{-p_3(x+h)}\right]$$

$$\frac{1}{k_c^2} q_2[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = -\frac{1}{k_c^2} p_3\left[De^{-p_3(x+h)}\right]$$

Substitute $k_c^2 = k_o^2 n_2^2 - k^2$, on the LHS and $k_c^2 = k_o^2 n_3^2 - k^2$ on the RHS gives

$$\frac{1}{k_o^2 n_2^2 - k^2} q_2[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = -\frac{1}{k_o^2 n_3^2 - k^2} p_3\left[De^{-p_3(x+h)}\right]_{x=-h}$$

Rearrange RHS

$$\frac{1}{k_o^2 n_2^2 - k^2} q_2[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = \frac{1}{k^2 - k_o^2 n_3^2} p_3\left[De^{-p_3(x+h)}\right]_{x=-h}$$

Substitute $q_2^2 = k_o^2 n_2^2 - k^2$ on the LHS and $p_3^2 = k^2 - k_o^2 n_3^2$ on the RHS

$$\frac{1}{q_2^2} q_2[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = \frac{1}{p_3^2} p_3\left[De^{-p_3(x+h)}\right]_{x=-h}$$

$$\frac{1}{q_2}[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = \frac{1}{p_3}\left[De^{-p_3(x+h)}\right]_{x=-h}$$

Substitute $D = -Asin(q_2 h) + Bcos(q_2 h)$

$$\frac{1}{q_2}[Acos(q_2 x) - Bsin(q_2 x)]_{x=-h} = \frac{1}{p_3}\left[\{-Asin(q_2 h) + Bcos(q_2 h)\}e^{-p_3(x+h)}\right]_{x=-h}$$

Substitute $A = \frac{q_2 B}{p_1}$

$$\frac{1}{q_2}\left[\left\{\frac{q_2 B}{p_1}\right\} cos(q_2 x) - Bsin(q_2 x)\right]_{x=-h} = \frac{1}{p_3}\left[\left\{-\left\{\frac{q_2 B}{p_1}\right\} sin(q_2 h) + Bcos(q_2 h)\right\} e^{-p_3(x+h)}\right]_{x=-h}$$

Divide both sides by B

$$\frac{1}{q_2}\left[\left\{\frac{q_2}{p_1}\right\} cos(q_2 x) - sin(q_2 x)\right]_{x=-h} = \frac{1}{p_3}\left[\left\{-\left\{\frac{q_2}{p_1}\right\} sin(q_2 h) + cos(q_2 h)\right\} e^{-p_3(x+h)}\right]_{x=-h}$$

Substitute x=-h

$$\frac{1}{q_2}\left[\left\{\frac{q_2}{p_1}\right\} cos(q_2 h) + sin(q_2 h)\right] \quad = \frac{1}{p_3}\left[\left\{-\left\{\frac{q_2}{p_1}\right\} sin(q_2 h) + cos(q_2 h)\right\}\right]$$

$$\left\{\frac{1}{p_1}\right\} cos(q_2 h) + \frac{1}{q_2} sin(q_2 h) = -\left\{\frac{q_2}{p_1 p_3}\right\} sin(q_2 h) + \frac{1}{p_3} cos(q_2 h)$$

$$\left(\frac{q_2}{p_1 p_3} - \frac{1}{q_2}\right) sin(q_2 h) = \left(\frac{1}{p_1} + \frac{1}{p_3}\right) cos(q_2 h) \qquad 5.29$$

$$\tan(q_2 h) = \frac{\left(\frac{1}{p_1} + \frac{1}{p_3}\right)}{\left(\frac{q_2}{p_1 p_3} - \frac{1}{q_2}\right)} = \frac{q_2(p_1 + p_3)}{(q_2^2 - p_1 p_3)} \qquad (5.30)$$

In general eq. 5.18 can be written cyclically as

$$\tan(q_2 h - m\pi) = \frac{q_2(p_1 + p_3)}{(q_2^2 - p_1 p_3)} \qquad (5.32)$$

where m = 0, 1, 2,… is the mode number. m stands for the number of times E_y crosses the x axis. In TE mode, the electric field is perpendicular to the direction of propagation.

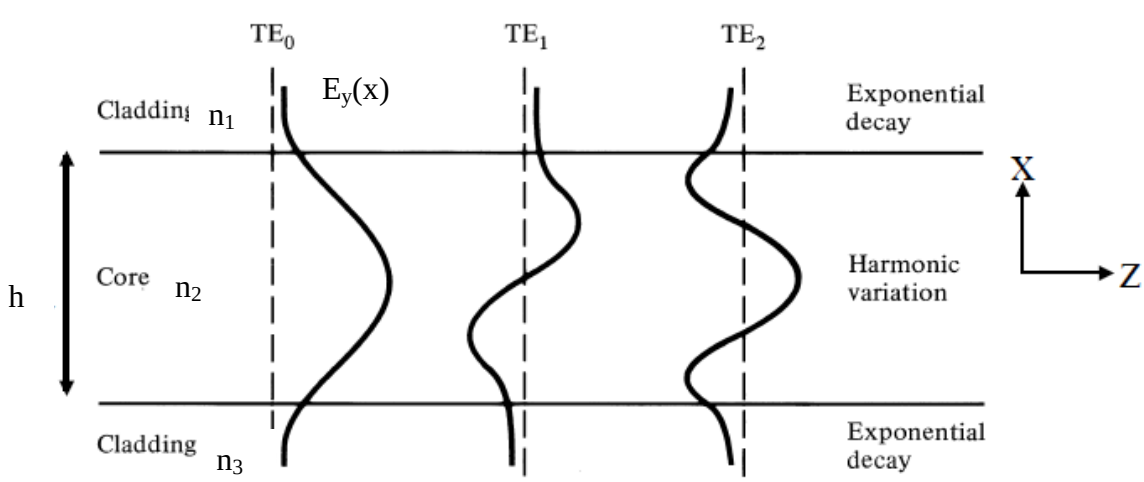

Fig. Field patterns for TE_0, TE_1 and TE_2 modes in dielectric slab waveguide.

Example 5.1

Find the cut-off widths of the first three TE and TM modes of a slab guide comprising air (n_1 = 1), GaAs (n_2 = 3.44) and GaAlAs (n_3 = 3.4) propagating an optical signal whose free-space wavelength is 1.15 μm.

Solution

We use $\tan(q_2 h_c - m\pi) = \frac{q_2(p_1 r_1 + p_3 r_3)}{q_2^2 - r_1 r_3 p_1 p_3}$

We know $p_3^2 = k^2 - k_o^2 n_3^2$. Hence cut-off occurs when $p_3 = 0$ for each mode which means k = $n_3 k_o$, and from $q_2^2 = k_o^2 n_2^2 - k^2$ we have

$$q_2^2 = k_o^2 n_2^2 - k^2 = k_o^2 n_2^2 - k_o^2 n_3^2 = k_o^2[n_2^2 - n_3^2] = \left(\frac{2\pi}{\lambda_o}\right)^2 [n_2^2 - n_3^2]$$

$$q_2^2 = \left(\frac{2\pi}{1.15\times10^{-6}}\right)^2 (3.44^2 - 3.4^2) \text{ or } q_2 = 2.86 \times 10^6/\text{m}$$

From $p_1^2 = k^2 - k_o^2 n_1^2 = k_o^2 n_3^2 - k_o^2 n_1^2 = k_o^2[n_3^2 - n_1^2] = \left(\frac{2\pi}{\lambda_o}\right)^2 [n_3^2 - n_1^2]$ we have

$$p_1^2 = \left(\frac{2\pi}{1.15\times10^{-6}}\right)^2 (3.44^2 - 1^2) \text{ or } p_1 = 1.18 \times 10^7/\text{m}$$

For TM modes $p_3 = 0$

$$\tan(q_2 h_c - m\pi) = \frac{q_2(p_1 r_1 + p_3 r_3)}{q_2^2 - r_1 r_3 p_1 p_3} = \frac{n_2^2 p_1}{n_1^2 q_2} = \frac{3.44^2 \times 1.18\times10^7}{1^2 \times 2.86\times10^6} = 73.5$$

$$q_2 h_c - m\pi = tan^{-1} 73.5 = 1.56$$

Thus for TM_0 m = 0 $\quad h_c = \frac{1.56}{2.86\times10^6} = 0.55$ μm

TM_1 m = 1 $\quad h_c = \frac{(1.56+\pi)}{2.86\times10^6} = 1.64$ μm

TM_2 m = 2 $\quad h_c = \frac{(1.56+2\pi)}{2.86\times10^6} = 2.74$ μm

For TE modes $p_3 = 0$

$$\tan(q_2 h - m\pi) = \frac{q_2(p_1 + p_3)}{(q_2^2 - p_1 p_3)} = \frac{p_1}{q_2} = \frac{1.18\times10^7}{2.86\times10^6} = 6.21$$

$$q_2 h_c - m\pi = tan^{-1}6.21 = 1.41$$

Thus for TE_0 m = 0 $h_c = \frac{1.41}{2.86\times10^6} = 0.49$ μm

TE_1 m = 1 $h_c = \frac{(1.41+\pi)}{2.86\times10^6} = 1.59$ μm

TE_2 m = 2 $h_c = \frac{(1.41+2\pi)}{2.86\times10^6} = 2.69$ μm

Example 5.2

Find the guiding layer thickness of an air/InGaAs/InP slab waveguide for which $n_1 = 1$, $n_2 = 3.23$ and $n_3 = 3.2053$, that gives

$$k = \frac{(n_2+n_3)k_o}{2}$$

for the fundamental TE mode at a free-space wavelength of 1.15 μm.

We use $\tan(q_2 h - mx) = \frac{q_2\,(p_1+p_3)}{(q_2^2-p_1p_3)}$

Recall

$$q_2^2 = k_o^2 n_2^2 - k^2 = \left(\frac{2\pi}{\lambda_o}\right)^2 n_2^2 - \left(\frac{(n_2+n_3)k_o}{2}\right)^2 = \left(\frac{2\pi}{\lambda_o}\right)^2 \left[n_2^2 - \left(\frac{(n_2+n_3)}{2}\right)^2\right]$$

$$q_2^2 == \left(\frac{2\pi}{1.55\times10^{-6}}\right)^2 \left[(3.23)^2 - \left(\frac{(3.23+3.2053)}{2}\right)^2\right]$$ or $q_2 = 1.4389\text{x}10^6\ \text{m}^{-1}$

$$p_1^2 = k^2 - k_o^2 n_1^2 = \left(\frac{(n_2+n_3)k_o}{2}\right)^2 - \left(\frac{2\pi}{\lambda_o}\right)^2 n_1^2 = \left(\frac{2\pi}{\lambda_o}\right)^2 \left[\left(\frac{(n_2+n_3)}{2}\right)^2 - n_1^2\right]$$

$$p_1^2 = \left(\frac{2\pi}{1.55\times10^{-6}}\right)^2 \left[\left(\frac{(3.23+3.2053)}{2}\right)^2 - (1)^2\right]$$ or $p_1 = 1.239738\text{x}10^7\ \text{m}^{-1}$

$$p_3^2 = k^2 - k_o^2 n_3^2 = \left(\frac{(n_2+n_3)k_o}{2}\right)^2 - \left(\frac{2\pi}{\lambda_o}\right)^2 n_3^2 = \left(\frac{2\pi}{\lambda_o}\right)^2 \left[\left(\frac{(n_2+n_3)}{2}\right)^2 - n_3^2\right]$$

$$p_3^2 = \left(\frac{2\pi}{1.55\times10^{-6}}\right)^2 \left[\left(\frac{(3.23+3.2053)}{2}\right)^2 - (3.2053)^2\right]$$ $p_3 = 1.14169\text{x}10^6\ \text{m}^{-1}$

Recall

$$\tan(q_2 h - mx) = \frac{q_2\,(p_1+p_3)}{(q_2^2-p_1p_3)}$$

[12]

For fundamental mode m=0 giving

$$\tan(q_2 h) = \frac{q_2(p_1+p_3)}{(q_2^2-p_1p_3)} = \frac{1.4389\text{x}10^6(1.239738\text{x}10^7+1.14169\text{xx}10^6)}{((1.4389\text{x}10^6)^2-1.239738\text{x}10^7 x 1.14169\text{xx}10^6)} = -1.20565$$

$$\tan(1.4389\text{x}10^6 h) = -1.20565 \qquad \text{h=1.978 μm}$$